SUBSEA DRONE AUTOPILOT

Autonomous Subsea Inspection & AI Fleet Command

DATABRICKS SOLUTION ACCELERATOR

Powered by Databricks Apps • Unity Catalog • Serverless

AI-POWERED AUTONOMOUS SUBSEA INSPECTION WITH COMPUTER VISION & FLEET COMMAND

Subsea Drone Autopilot delivers an end-to-end AI-powered system for controlling a fleet of 5 underwater ROV/AUV drones — from mission planning and real-time computer vision inspection through defect analysis and maintenance recommendations.

BUSINESS VALUE

- **Reduce inspection costs** from $50K–$200K per campaign with autonomous AI-planned missions
- **Detect subsea defects automatically** with YOLOv8-style CV — corrosion, cracks, marine growth, coating damage
- **Improve safety** by replacing manual ROV operations with autonomous drone dispatch and monitoring
- **Accelerate reporting** with AI-generated inspection reports including defect cards and evidence frames
- **Enable knowledge-driven operations** with RAG-powered Q&A over subsea manuals and fleet history

CHALLENGES ADDRESSED

- **Expensive manual inspections** — ROV campaigns costing $50K–$200K with slow turnaround
- **Human error in defect detection** — manual frame review missing subtle corrosion and coating damage
- **No autonomous planning** — mission routing and drone selection done manually without safety optimization
- **Fragmented inspection data** — reports, frames, and telemetry in disconnected systems
- **Limited knowledge access** — subsea manuals and standards not integrated into workflows

WHO IT'S FOR

Subsea Engineer
Mission planning, live CV inspection viewer, and defect analysis reports

Integrity Manager
Fleet digital twin, maintenance advisor, and condition-based recommendations

Operations Coordinator
Fleet command dashboard, ocean operations map, and mission history

Knowledge Analyst
RAG-powered Q&A for subsea manuals, fleet status, and inspection history

5	4	7	24
Drones	AI Agents	Dashboard Pages	CV Frames

QUICK START GUIDE

1. **Clone Repository** — Clone the Subsea Drone Autopilot repo to your machine
2. **Run DDL** — Create Delta tables on your SQL warehouse
3. **Build & Deploy** — npm build, create deploy package, databricks apps deploy
4. **Grant Permissions** — Give app service principal catalog access

GETTING THE MOST OUT OF SUBSEA DRONE AUTOPILOT

Plan Missions
Use the Autopilot Agent to dispatch inspection missions with safety-enforced drone selection

Watch Live Inspection
Stream real-time CV-annotated frames with defect bounding boxes and severity scores

Review Reports
Analyze mission analytics with defect cards, evidence frames, and prioritized actions

Ask Knowledge Agent
Query subsea manuals, fleet data, and inspection history with source citations

TARGET INDUSTRY

Offshore Oil & Gas — Subsea infrastructure inspection for FPSO vessels, risers, manifolds, flowlines, and mooring systems. Applicable to any offshore operator managing subsea assets.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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